Project Design Phase-II
Data Flow Diagram & User Stories

Date	01 October 2026
Team ID	SWTID-2026-6675
Project Name	Auto Ticket Classification Using Flow Designer
Maximum Marks	4 Marks

Data Flow Diagrams:

The Data Flow Diagram (DFD) represents the flow of incident information within the Auto Ticket Classification Using Flow Designer system. The system is designed to automate the classification of IT support incidents submitted by students and teachers in a school environment.

Students and teachers submit IT-related incident requests through the ServiceNow platform. These requests may include common problems such as Wi-Fi connectivity issues, projector failures, password problems, and slow computer performance. The incident contains important information such as the Short Description and Description, which helps the system understand the reported issue.

When a new incident is created, the incident details are stored in the Incident Records. The creation of a new incident triggers the configured Flow Designer workflow. The workflow retrieves the relevant information from the incident record and begins the classification process.

Flow Designer analyzes the Short Description and Description using predefined keywords and conditions. For example, keywords related to Wi-Fi or network connectivity can indicate a network-related issue, while words related to passwords, projectors, or computer performance can indicate their respective issue types.

After analyzing the incident information, the system determines the appropriate category and subcategory for the ticket. The classification result is automatically updated in the incident record. This reduces the need for IT support staff to manually review and categorize every incoming ticket.

The classified incident is then available to the IT support team for further investigation and resolution. The support team can review the incident details and provide the required technical support.

Overall, the data flow begins with incident submission, followed by incident storage, keyword analysis, automatic classification, incident record update, and IT support resolution. The proposed system reduces repetitive manual work, improves classification consistency, and helps the school IT helpdesk process incident requests more efficiently.

Example:

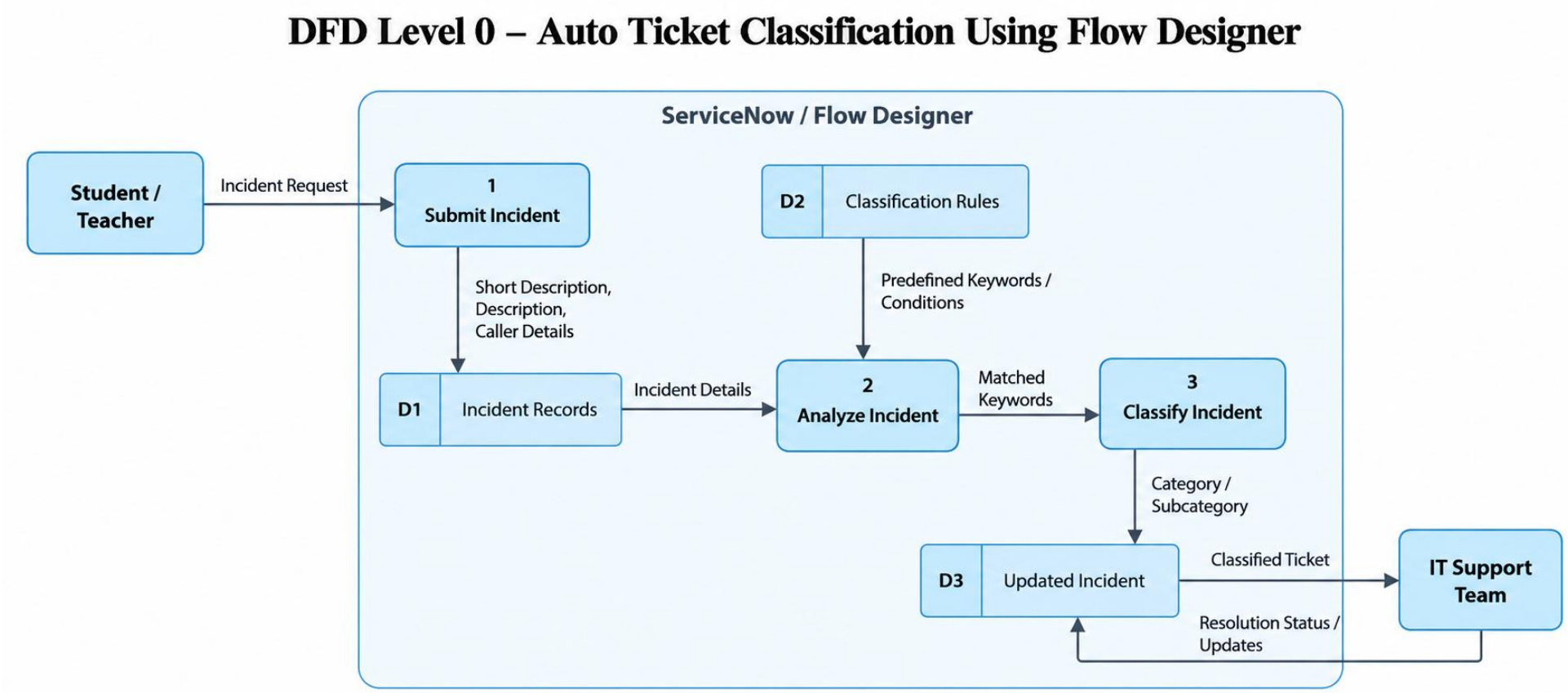


Figure: Data Flow Diagram of the Auto Ticket Classification Using Flow Designer system.

User Stories:

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Student / Teacher	Incident Submission	USN-1	As a user, I can submit an IT incident by providing the required incident details.	Incident is successfully created with the provided details.	High	Sprint-1
Student / Teacher	Incident Submission	USN-2	As a user, I can enter the issue in the Short Description field.	Short Description is saved with the incident.	High	Sprint-1
Student / Teacher	Incident Submission	USN-3	As a user, I can provide additional issue details in the Description field.	Description is saved correctly with the incident.	High	Sprint-1
IT Support Team	Ticket Classification	USN-4	As an IT support member, I want newly created incidents to be automatically analyzed.	Flow Designer is triggered when an incident is created.	High	Sprint-2
IT Support Team	Ticket Classification	USN-5	As an IT support member, I want the system to identify keywords from the incident details.	Relevant predefined keywords are identified from Short Description and Description.	High	Sprint-3
IT Support Team	Ticket Classification	USN-6	As an IT support member, I want incidents to be automatically classified based on predefined conditions.	Appropriate category is assigned according to the matching condition.	High	Sprint-3

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
IT Support Team	Category & Subcategory	USN-7	As an IT support member, I want the appropriate subcategory to be selected for a classified incident.	Correct subcategory is assigned based on the selected category and rule.	High	Sprint-3
IT Support Team	Ticket Validation	USN-8	As an IT support member, I want to verify the classification result.	The incident record displays the expected category and subcategory.	Medium	Sprint-4
Administrator	Flow Configuration	USN-9	As an administrator, I can modify the predefined classification conditions when required.	Classification rules can be updated and saved successfully.	Medium	Sprint-5
Administrator	Deployment	USN-10	As an administrator, I can complete the Update Set after configuration and testing.	Update Set is successfully moved to the completed state.	Medium	Sprint-5